

Question:

Write a JAVA program (**StudentUtilityProgram**) to demonstrate the basics of Java: methods with parameters and return values, conditional statements (if-else, switch, ternary operator) and loops (for, do-while). Create a menu-driven program in which each menu option calls a separate method.

Execution Steps:

Step 1: Program execution starts from main().

Step 2: Scanner object sc is created.

Step 3: The do-while loop starts and the menu is displayed.

Step 4: User enters 2.

Step 5: choice = sc.nextInt() stores 2 in choice.

Step 6: switch(choice) matches case 2.

Step 7: The prompt "Enter number:" is displayed and the user enters 7.

Step 8: evenOdd(7) is called.

Step 9: The condition $7 \% 2 == 0$ is false, so the ternary operator selects the Odd message.

Step 10: "7 is Odd" is displayed and break leaves the switch.

Step 11: The loop condition (choice != 0) is true, so the menu is displayed again.

Step 12: User enters 0. case 0 prints "Exiting program...".

Step 13: The loop condition (choice != 0) is false and the loop ends.

Step 14: Scanner is closed.

Step 15: Program ends.

Algorithm:

Step 1: Start.

Step 2: Create a Scanner object to read input.

Step 3: Display the menu (options 1 to 13 and 0 to exit).

Step 4: Read the user's choice.

Step 5: Use switch to check the choice:

1 - read name, roll number, 3 marks; call displayInfo() (total, percentage, grade)

2 - call evenOdd() 3 - call findLargest() 4 - call displayResult()

5 - call dayOfWeek() 6 - call printTable() 7 - call displayNumbers()

8 - call sumNaturalNumbers() 9 - call fibonacciSeries() 10 - call factorial()

11 - call isPrime() 12 - call maxOfTwo() 13 - call areaOfCircle()

0 - display "Exiting program..."

Step 6: If the choice is not valid, display "Invalid choice!".

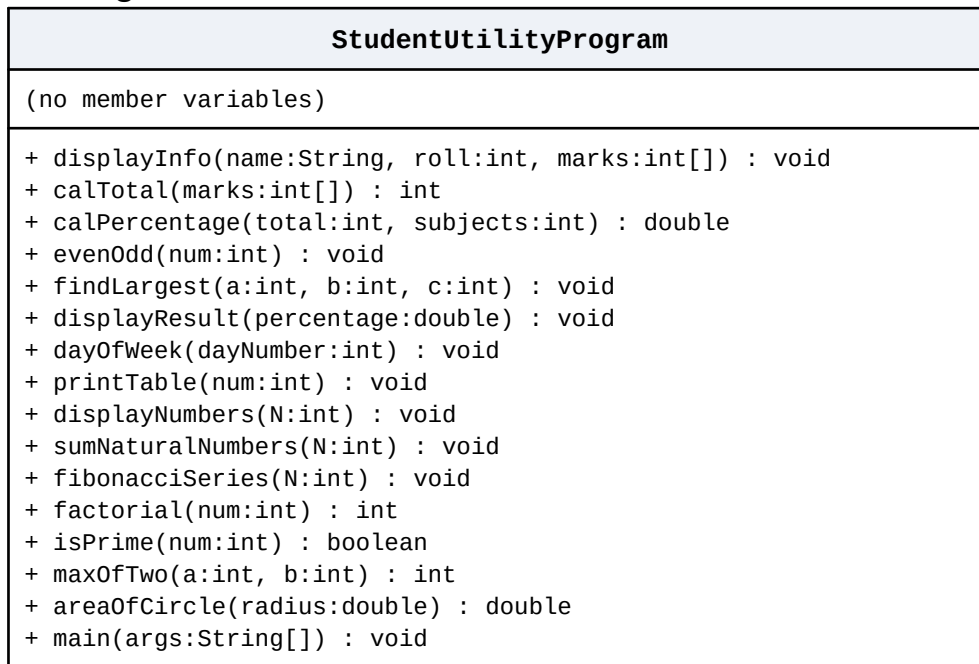
Step 7: Display the result returned or printed by the called method.

Step 8: Repeat Steps 3 to 7 while choice is not 0.

Step 9: Close the Scanner.

Step 10: Stop.

UML Class Diagram:



All methods are public static, so they are called directly from main() without creating an object.

Code:

```
package studentutility;
import java.util.Scanner;

public class StudentUtilityProgram {
    // Part A: basics
    public static void displayInfo(String name, int roll,int[] marks) {
        int total = calTotal(marks);
        double percentage = calPercentage(total, marks.length);
        System.out.println("\nStudent information : ");
        System.out.println("Name of the student: " + name);
        System.out.println("Roll number: " + roll);
        System.out.println("Total marks: " + total);
        System.out.println("Percentage: " + percentage );
        displayResult(percentage);
    }
    public static int calTotal(int[] marks) {
        int total = 0;
        for(int m : marks)
            total += m;
        return total;
    }
    public static double calPercentage( int total, int subjects) {
        return (double) total / subjects;
    }
    public static void evenOdd(int num) {
        System.out.println(num % 2 == 0 ? num + " is Even" : num + " is Odd");
    }
    public static void findLargest(int a, int b, int c) {
        int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
        System.out.println("Largest number is: " + largest);
    }
    public static void displayResult(double percentage) {
        if (percentage >= 90) System.out.println("Grade: A+");
        else if (percentage >= 75) System.out.println("Grade: A");
        else if (percentage >= 60) System.out.println("Grade: B");
        else if (percentage >= 40) System.out.println("Grade: C");
        else System.out.println("Grade: Fail");
    }
    public static void dayOfWeek(int dayNumber) {
```

```

        switch (dayNumber) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
            case 3: System.out.println("Wednesday"); break;
            case 4: System.out.println("Thursday"); break;
            case 5: System.out.println("Friday"); break;
            case 6: System.out.println("Saturday"); break;
            case 7: System.out.println("Sunday"); break;
            default: System.out.println("Invalid day number!");
        }
    }

    public static void printTable(int num) {
        System.out.println("Multiplication Table of " + num);
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " x " + i + " = " + (num * i));
        }
    }

    public static void displayNumbers(int N) {
        for (int i = 1; i <= N; i++) System.out.print(i + " ");
        System.out.println();
    }

    public static void sumNaturalNumbers(int N) {
        int sum = 0;
        for (int i = 1; i <= N; i++) sum += i;
        System.out.println("Sum of first " + N + " natural numbers = " + sum);
    }

    public static void fibonacciSeries(int N) {
        int a = 0, b = 1;
        System.out.print("Fibonacci Series: ");
        for (int i = 1; i <= N; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
        System.out.println();
    }

    public static int factorial(int num) {
        int fact = 1;
        for (int i = 1; i <= num; i++) fact *= i;
        return fact;
    }

    public static boolean isPrime(int num) {
        if (num <= 1) return false;
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) return false;
        }
        return true;
    }

    public static int maxOfTwo(int a, int b) {
        return (a > b) ? a : b;
    }

    public static double areaOfCircle(double radius) {
        return Math.PI * radius * radius;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("\n--- Student Utility Program ---");
            System.out.println("1. Student Info");
            System.out.println("2. Even/Odd Check");
            System.out.println("3. Largest of Three Numbers");
            System.out.println("4. Display Grade");
            System.out.println("5. Day of Week");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Display Numbers 1 to N");
            System.out.println("8. Sum of Natural Numbers");
            System.out.println("9. Fibonacci Series");

```

```

System.out.println("10. Factorial");
System.out.println("11. Prime Check");
System.out.println("12. Max of Two Numbers");
System.out.println("13. Area of Circle");
System.out.println("0. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch (choice) {
case 1:
    sc.nextLine();
    System.out.print("Enter Name: ");
    String name = sc.nextLine();
    System.out.print("Enter Roll Number: ");
    int roll = sc.nextInt();
    int[] marks = new int[3];
    System.out.println("Enter 3 subject marks:");
    for (int i = 0; i < 3; i++) marks[i] = sc.nextInt();
    displayInfo(name, roll, marks);
    break;
case 2:
    System.out.print("Enter number: ");
    evenOdd(sc.nextInt());
    break;
case 3:
    System.out.print("Enter three numbers: ");
    findLargest(sc.nextInt(), sc.nextInt(), sc.nextInt());
    break;
case 4:
    System.out.print("Enter percentage: ");
    displayResult(sc.nextDouble());
    break;
case 5:
    System.out.print("Enter day number (1-7): ");
    dayOfWeek(sc.nextInt());
    break;
case 6:
    System.out.print("Enter number: ");
    printTable(sc.nextInt());
    break;
case 7:
    System.out.print("Enter N: ");
    displayNumbers(sc.nextInt());
    break;
case 8:
    System.out.print("Enter N: ");
    sumNaturalNumbers(sc.nextInt());
    break;
case 9:
    System.out.print("Enter N: ");
    fibonacciSeries(sc.nextInt());
    break;
case 10:
    System.out.print("Enter number: ");
    int num = sc.nextInt();
    System.out.println("Factorial = " + factorial(num));
    break;
case 11:
    System.out.print("Enter number: ");
    int primeNum = sc.nextInt();
    System.out.println(primeNum + (isPrime(primeNum) ? " is Prime" : " is Not Prime"));
    break;
case 12:
    System.out.print("Enter two numbers: ");
    System.out.println("Max = " + maxOfTwo(sc.nextInt(), sc.nextInt()));
    break;
case 13:
    System.out.print("Enter radius: ");
    System.out.println("Area = " + areaOfCircle(sc.nextDouble()));
    break;
}

```

```
        case 0:
            System.out.println("Exiting program...");
            break;
        default:
            System.out.println("Invalid choice!");
    }
} while (choice != 0);
sc.close();
}}
```

Output:

```
--- Student Utility Program ---
1. Student Info
2. Even/Odd Check
3. Largest of Three Numbers
4. Display Grade
5. Day of Week
6. Multiplication Table
7. Display Numbers 1 to N
8. Sum of Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Check
12. Max of Two Numbers
13. Area of Circle
0. Exit
Enter your choice: 1
Enter Name: Shubhang Chourey
Enter Roll Number: 102
Enter 3 subject marks:
85 90 78
Student information :
Name of the student: Shubhang Chourey
Roll number: 102
Total marks: 253
Percentage: 84.33333333333333
Grade: A
... (menu displayed again) ...
Enter your choice: 2
Enter number: 7
7 is Odd
... (menu displayed again) ...
Enter your choice: 3
Enter three numbers: 12 45 30
Largest number is: 45
... (menu displayed again) ...
Enter your choice: 4
Enter percentage: 82.5
Grade: A
... (menu displayed again) ...
Enter your choice: 5
Enter day number (1-7): 3
Wednesday
... (menu displayed again) ...
Enter your choice: 10
Enter number: 5
Factorial = 120
... (menu displayed again) ...
Enter your choice: 11
Enter number: 17
17 is Prime
... (menu displayed again) ...
Enter your choice: 13
Enter radius: 5
Area = 78.53981633974483
... (menu displayed again) ...
Enter your choice: 0
Exiting program...
```

Observation Questions:

Question 1: Explain the structure of a Java program.

Package declaration: `package studentutility;`

Import statements: `import java.util.Scanner;`

Class definition: `public class StudentUtilityProgram`

main method: `public static void main(String[] args)`

Methods: user-defined methods for tasks (factorial, prime check, etc.)

So the structure is: Package → Imports → Class → main() → Methods.

Question 2: Differentiate between primitive and non-primitive data types.

Primitive: built-in basic types, stored directly in memory, keywords are always lowercase. Examples: `int`, `char`, `double`, `boolean`.

Non-primitive: created by the programmer or provided by the Java library. Examples: `String`, `Array`, `Class`, `Object`.

Question 3: Compare if-else and switch.

if-else: used for complex conditions; works with relational operators (`<`, `>`, `==`, etc.); flexible but can become lengthy.

switch: used when a variable is compared against fixed values; works with `int`, `char`, `String`, `enums`; cleaner for multiple discrete choices.

Question 4: Compare the loops in Java.

for loop: used when the number of iterations is known.

while loop: used when the condition must be checked before each iteration.

do-while loop: executes at least once; the condition is checked after the iteration.